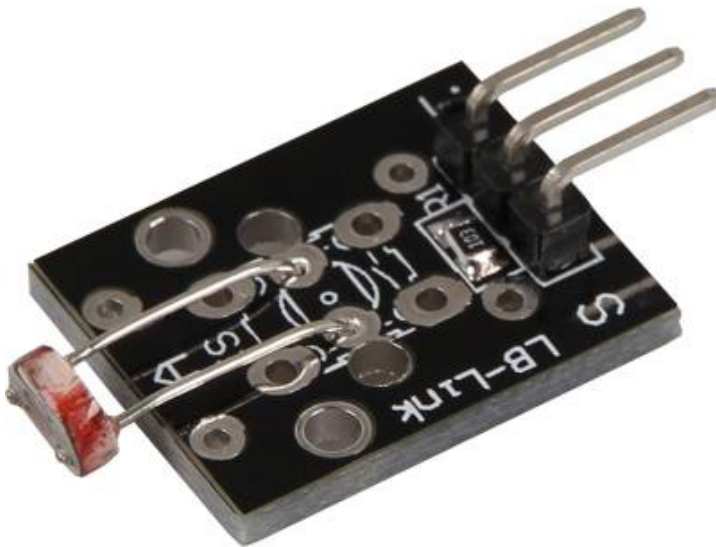


KY-018 Photoresistor module

Contents

1 Picture	1
2 Technical data / Short description	1
3 Pinout	2
4 Code example Arduino	2
5 Code example Raspberry Pi	3

Picture

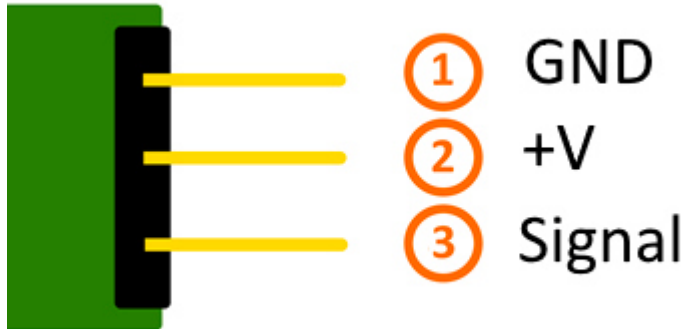


Technical data / Short description

Includes a LDR-resistor (Light dependent resistor), which decreases its resistance at brighter surrounding.

You can determine the resistance of the resistor by using a voltage divider, where a known voltage will be divided between a known and an unknown variable resistor. You can calculate the resistance with the measured voltage - the code example below includes the exact calculation.

Pinout



Code example Arduino

The example will measure the current sensors voltage and calculates the current resistance of the sensor.

```
int sensorPin = A5; // Declaration of the input pin

// Serial output in 9600 Baud
void setup()
{
    Serial.begin(9600);
}

// The program measures the current voltage at the sensor ,
// takes the value of the known resistor and calculates the current resistance of the sensor.
// After that it show the result via serial output.

void loop()
{
    // Current voltage will be measured...
    int rawValue = analogRead(sensorPin);
    float voltage = rawValue * (5.0/1023) * 1000;

    float resitance = 10000 * ( voltage / ( 5000.0 - voltage) );

    // and here it will be outputted via serial infterface
    Serial.print("Voltage value:");      Serial.print(voltage); Serial.print("mV");
    Serial.print(", Resistor value:"); Serial.print(resitance); Serial.println("Ohm");
    Serial.println("-----");

    delay(500);
}
```

Connections Arduino:

Sensor GND	= [Pin GND]
Sensor +V	= [Pin 5V]
Sensor Signal	= [Pin A5]

Example program download

[Single_Analog_Sensor](#)

Code example Raspberry Pi

!! Attention !! Analog Sensor !! Attention !!

Unlike the Arduino, the Raspberry Pi doesn't provide an ADC (Analog Digital Converter) on its Chip. This limits the Raspberry Pi if you want to use a non digital Sensor.

To evade this, use our *Sensorkit X40* with the *KY-053* module, which provides a 16 Bit ADC, which can be used with the Raspberry Pi, to upgrade it with 4 additional analog input pins. This module is connected via I2C to the Raspberry Pi.

It measures the analog data and converts it into a digital signal which is suitable for the Raspberry Pi.

So we recommend to use the KY-053 ADC if you want to use analog sensors along with the Raspberry Pi.

For more information please look at the infosite: [\[\[KY-053 Analog Digital Converter\]\]](#)

!! Attention !! Analog Sensor !! Attention !!

The program uses the specific ADS1x15 and I2C python-libraries from the company Adafruit to control the ADS1115 ADC. You can find these here: [\[https://github.com/adafruit/Adafruit-Raspberry-Pi-Python-Code\]](https://github.com/adafruit/Adafruit-Raspberry-Pi-Python-Code) published under the BSD-License [\[Link\]](#). You can find the needed libraries in the lower download package.

The program measures the current voltage at the sensor, takes the value of the known resistor and calculates with it the current resistance of the sensor.

After that it will show the result via serial output.

```
#####
### Copyright by Joy-IT
### Published under Creative Commons Attribution-NonCommercial-ShareAlike 3.0 Unported License
### Commercial use only after permission is requested and granted
###
### KY-053 Analog Digital Converter - Raspberry Pi Python Code Example
###
#####

# This code is using the ADS1115 and the I2C Python Library for Raspberry Pi
# This was published on the following link under the BSD license
# [https://github.com/adafruit/Adafruit-Raspberry-Pi-Python-Code]
from Adafruit_ADS1x15 import ADS1x15
from time import sleep

# import needed modules
import time, signal, sys, os
import RPi.GPIO as GPIO
GPIO.setmode(GPIO.BCM)
GPIO.setwarnings(False)

# initialise variables
delayTime = 0.5 # in Sekunden

# assigning the ADS1x15 ADC
```

KY-018 Photoresistor module

```

ADS1015 = 0x00 # 12-bit ADC
ADS1115 = 0x01 # 16-bit

# choosing the amplifying gain
gain = 4096 # +/- 4.096V
# gain = 2048 # +/- 2.048V
# gain = 1024 # +/- 1.024V
# gain = 512 # +/- 0.512V
# gain = 256 # +/- 0.256V

# choosing the sampling rate
# sps = 8 # 8 Samples per second
# sps = 16 # 16 Samples per second
# sps = 32 # 32 Samples per second
sps = 64 # 64 Samples per second
# sps = 128 # 128 Samples per second
# sps = 250 # 250 Samples per second
# sps = 475 # 475 Samples per second
# sps = 860 # 860 Samples per second

# assigning the ADC-Channel (1-4)
adc_channel_0 = 0 # Channel 0
adc_channel_1 = 1 # Channel 1
adc_channel_2 = 2 # Channel 2
adc_channel_3 = 3 # Channel 3

# initialise ADC (ADS1115)
adc = ADS1x15(ic=ADS1115)

#####

# #####
# Main Loop
# #####
# Reading the values from the input pins and print to console

try:
    while True:
        #read values
        adc0 = adc.readADCSingleEnded(adc_channel_0, gain, sps)
        adc1 = adc.readADCSingleEnded(adc_channel_1, gain, sps)
        adc2 = adc.readADCSingleEnded(adc_channel_2, gain, sps)
        adc3 = adc.readADCSingleEnded(adc_channel_3, gain, sps)

        # print to console
        print "Channel 0:", adc0, "mV "
        print "Channel 1:", adc1, "mV "
        print "Channel 2:", adc2, "mV "
        print "Channel 3:", adc3, "mV "
        print "-----"

        time.sleep(delayTime)

```

KY-018 Photoresistor module

Analog Signal = Analog 0 [Pin A0 (ADS1115 - KY-053)]

ADS1115 - KY-053:

VDD	= 3,3V	[Pin 17]
GND	= GND	[Pin 09]
SCL	= GPIO03 / SCL	[Pin 05]
SDA	= GPIO02 / SDA	[Pin 03]
A0	= look above	[Sensor: analog signal]

Example program download

[KY-053_RPi_AnalogDigitalConverter](#)

To start, enter the command:

```
sudo python RPi_Single_Analog_Sensor.py
```